PP-HFM: Patient-Private Hierarchical Federated Multimodal Learning

Project report: the idea, the system, every experiment and what each one showed

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MIT Bengaluru (MAHE) Status as of 28 October 2026

This is an internal report, not the paper. Every number is taken from the result files under `SIMULATORS/results/`; results still being re-run are marked pending.

1 The project in one page

The one-sentence version. We built a three-tier federated learning system (devices → edge servers → cloud) that trains a multimodal clinical model without moving raw data, and that protects every *patient*, not just every data *record*, with a mathematically proven differential-privacy guarantee. We then measured, carefully, what that protection costs.

The problem. Hospitals and IoT devices produce data that cannot leave the site. Federated learning (FL) solves this: each site trains locally and only sends model updates. Differential privacy (DP) adds noise so that the updates do not reveal any individual. But almost every published DP system protects a *record* (one 10-minute window of sensor data). A patient contributes many records, around 8 in VitalDB and 28 in eICU. Protecting each record does *not* protect the patient: if you convert after the fact ("group privacy"), a guarantee of $\varepsilon \approx 1$ per record becomes $\varepsilon \approx 8$ per patient, which is almost no protection.

Our answer. PP-HFM makes the patient the privacy unit from the start. All of a patient's records are summed into one gradient, that sum is clipped, and noise is added to it. Each client (hospital) gets its own noise level calibrated to its own patient count. Every release in the hierarchy (devices, edges, cloud) is accounted for, and a theorem proves the guarantee.

What we found, in brief.

1. **The system works.** Uplink traffic falls 90.8–96.1 %, energy per round falls 72.3–79.6 %, and inference takes at most 2.3 ms (target was 100 ms), on all three datasets.
2. **Patient-level privacy at $\varepsilon = 0.9$ costs no measurable accuracy on VitalDB** but costs 3.1 AUROC points on eICU, because of the smallest hospitals.
3. **The thing that makes strict privacy survivable is pretrained initialization**, not the trick of privatizing only the model head. A randomly initialized private model loses 0.19 AUROC; a pretrained one loses 0.004.
4. **The cost of protecting patients grows with the number of records per patient, k, and then flattens.** This follows the curve $A(1 - 1/k)$, and vanishes at $k = 1$, where a patient *is* a record. It was forecast in advance and tested on a second hospital corpus: the *shape* transferred, the *size* did not.
5. Several things we expected did *not* hold, and we report them: the membership-inference attack did not find the big leakage difference we predicted; private conformal prediction gave useless prediction sets; fusion did not help on eICU.

2 How the project progressed (phases)

Phase	What happened
1–2	Literature review and problem definition. Four research gaps identified (Section 2.1); four quantified objectives fixed.
3	Environment: Python stack, NS-3 network simulator (v3.46.1), CloudSim Plus (v8.5.6) for energy, GPU utilities.
4	Both simulators implemented: network (latency, traffic, reliability) and edge–cloud resource/energy model. First system-level numbers.
5	The learning system: multimodal model, hierarchical FL, DP, compression, conformal calibration. All 17 deliverables on VitalDB and WEAR (5 Aug). Four silent bugs found and fixed.
6	Ablations and baselines: which component does what. Hierarchy resolved at 6 seeds; pretrained-initialization finding.
7	External validation on eICU (187 hospitals), the k-scaling law, pre-registered tests, paper writing.
Fix (22 Oct–)	A flaw in the patient-DP mechanism was found while writing the proof. Fixed, and every patient-level experiment is being re-run in stages (Section 6).

2.1 The four research gaps

1. **No multimodal fusion in private FL**: existing systems treat data as generic gradients.
2. **Privacy vs. efficiency unresolved**: HED-FL has a hierarchy but no DP; FL2DP has DP but no hierarchy.
3. **Real-time QoS** ($< 100\,\text{ms}$) is not addressed in robust FL frameworks.
4. **The privacy unit**: guarantees are per record, while clinical deployments must protect people.

2.2 The four objectives

Objective	Target	Measured by
O1 Uplink traffic reduction	$\geq 50\,\%$	NS-3 packet-level simulation
O2a Privacy budget	$\varepsilon \leq 1$	privacy accountant
O2b Utility loss from privacy	≤ 2 points	paired training runs
O3a Inference latency (P95)	$< 100\,\text{ms}$	ONNX Runtime, CPU
O3b Update delivery	$> 99\,\%$	NS-3
O4 Energy saving per round	$\geq 20\,\%$	CloudSim Plus

Each objective is measured by exactly one tool, so no number is reported twice from two sources.

3 The system

3.1 The three tiers

Tier 1 – clients (devices / hospitals). Each holds its own patients. It trains the model locally with patient-level DP-SGD (below) and sends a compressed update: only the top 20 % of values, quantized to 8 bits, with error feedback so nothing is permanently lost.

Tier 2 – edge servers. Each edge averages the updates of its clients, $\tau_e = 5$ times per cloud round. Most traffic therefore stays local and never crosses the wide-area link.

Tier 3 – cloud. Averages the edges once per round ($R = 10$ rounds) and broadcasts the global model.

3.2 The model

Two encoders, one per modality (e.g. vital-sign time series and pre-operative clinical variables), whose embeddings are concatenated and passed to a small *fusion head*. The model has 1.86 M

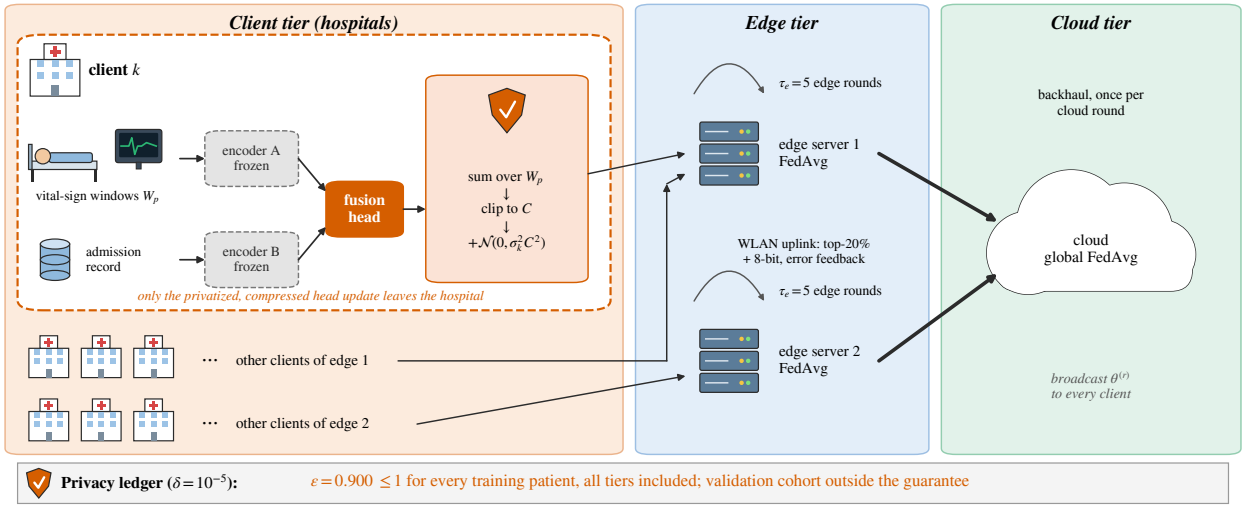


Figure 1: PP-HFM: clients train with patient-level DP, edges aggregate several rounds locally, the cloud aggregates the edges.

parameters; the head is 0.56 M (30 %). The encoders are pretrained once and frozen; federated training updates only the head. All normalization layers are group/layer norm, because batch norm mixes patients inside a batch and would break the per-patient privacy accounting.

3.3 Patient-level DP-SGD (the core mechanism)

1. **Sample patients, not records.** Each patient is included in a step independently with probability q (Poisson sampling).
2. **Sum per patient, then clip.** All of a sampled patient's record gradients are summed into one vector, and that vector is clipped to norm $C = 1$. One patient can therefore move the update by at most C, however many records they have.
3. **Add Gaussian noise** of scale $\sigma_k C$ and divide by the *expected* batch size $q|P_k|$ (not the realized one; see Section 6).
4. **Per-client noise.** Every patient belongs to exactly one client, so each client's noise σ_k can be calibrated to its own population (parallel composition), instead of forcing the worst hospital's noise on everyone. On eICU this means a median σ of 13.4 instead of 29.3 everywhere.
5. **Accounting.** The PRV accountant computes the total $(\varepsilon, \delta) = (0.9, 10^{-5})$ over all rounds and tiers.

Theorem 1 (what is proven). For declared public cohort sizes and fixed preprocessing, the full training transcript of every tier is (ε, δ)-DP with respect to adding or removing one patient. It covers the *training* process. It is end-to-end only if the encoders and the standardization constants are also fitted on public data (Section 5.15).

4 The data

	VitalDB	WEAR	eICU
Domain	surgery	wearable activity	intensive care
Clients	9 surgical specialties	22 participants	187 hospitals
Individuals	6,257 cases	22	132,736
Records per individual ($\bar{k}$)	7.8	$\approx$ 1,750	28.5
Modality A	9 vital trends + masks	inertial sensors	6 vitals + masks
Modality B	34 pre-op variables	2048-d video features	5 admission values
Task	ICU admission	18 activities	mortality
Positive rate	19.65 %	–	4.22 %

Three choices protect the results from silent inflation: splits are by individual (never by window);

only pre-operative variables are used (the intra-operative ones encode the outcome); missing values get an explicit indicator channel (an arterial line is placed only in patients judged at risk, so "missing" is itself informative).
On eICU a pool of 526,867 windows from individuals who are in no client is reserved for pretraining the encoders, so eICU encoders never see protected patients. On VitalDB the cohort is too small for that, so the headline uses "oracle" encoders pretrained on client data, which is an upper bound (the cost of doing it properly is measured in Section 5.16).

5 Experiments, one by one

Each experiment below has the same four parts: the question, what we did, the result, and what it means. "Points" means percentage points of AUROC (0.01 AUROC = 1 point). Intervals are 95 % confidence intervals across seeds.

5.1 System objectives on three corpora

Question. Does the full system meet the four objectives?
What we did. Ran the complete system on VitalDB, WEAR and eICU; network measured in NS-3 at packet level on every round, energy in CloudSim Plus, latency with ONNX Runtime on CPU (1,000 runs after 100 warm-up).
Result.

	VitalDB	WEAR	eICU	Target
O1 Uplink reduction	90.8 %	96.1 %	92.3 %	≥ 50 %
O1b Round-latency cut	16.9 %	21.7 %	33.4 %	> 0
O2a Budget ε	0.900	0.766 (record)	0.900	≤ 1
O2b Utility loss [pts]	−0.90	1.97 (inconcl.)	3.09 (**not met**)	≤ 2
O3a P95 latency	2.30 ms	0.75 ms	0.24 ms	< 100 ms
O3b Update delivery	100.00 %	99.94 %	97.83 % (**not met**)	> 99 %
O4 Energy saving	72.3 %	76.8 %	79.6 %	≥ 20 %

What it means. Communication, latency and energy targets are met everywhere by a wide margin. The two misses are both on eICU and both have a known cause: utility (small hospitals, next experiments) and delivery (187 clients sharing two edges, Section 5.3).

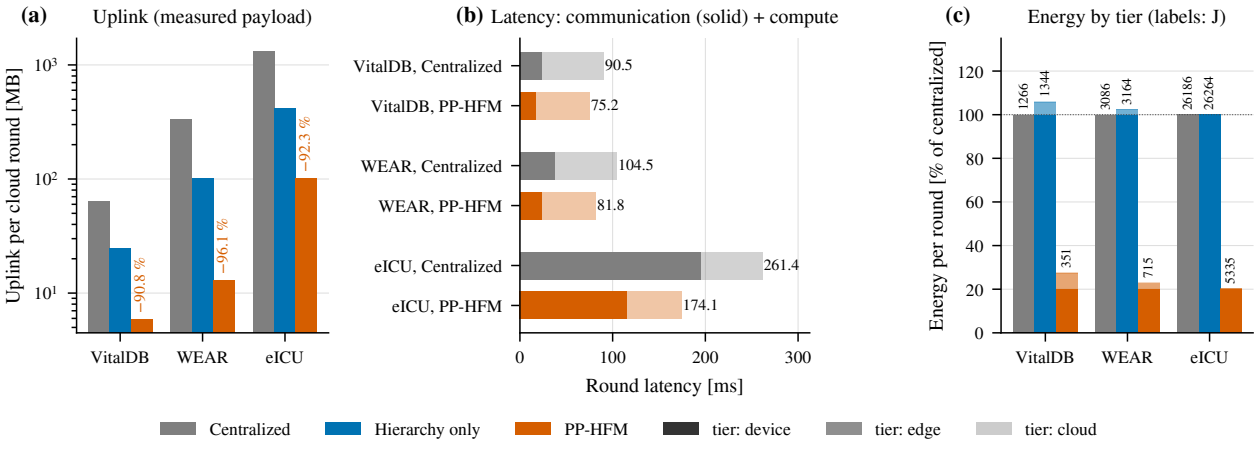


Figure 2: System cost relative to centralized training: uplink, round latency, energy by tier.

5.2 Inference latency

Question. Can the model answer within 100 ms at the edge?
What we did. Exported to ONNX, timed on a CPU; also tried 8-bit quantization.

Result. P95 latency 2.30 / 0.75 / 0.24 ms. On VitalDB 94 % of it is the forward pass. 8-bit quantization made it *slower* (P95 17.3 ms) and broke the WEAR GRU, so it was rejected.
What it means. More than a fortyfold margin. Latency is not a constraint for this model.

5.3 Update delivery vs. number of edges (eICU)

Question. Why does eICU miss the 99 % delivery target?
What we did. Kept everything fixed and varied the number of edge servers.
Result. With 2 edges, 187 clients contend for the wireless medium: 97.83 % (centralized: 95.11 %). Delivery rises monotonically, meets 99 % on average from 6 edges and in every round from 8 (about 23 clients per edge), reaching 99.94 % at 16.
What it means. It is a provisioning question, not a design flaw: the hierarchy needs enough edges for the client count.

5.4 Utility cost of privacy (O2b)

Question. How much accuracy does $\varepsilon = 0.9$ patient-level privacy cost?
What we did. Identical system with and without DP, 5 paired seeds, per-client noise, PRV accountant. AUROC is primary, because at the default 0.5 threshold the private model predicts the majority class on every seed (accuracy = base rate), so accuracy measures the threshold, not the model.
Result.

- **VitalDB:** AUROC 0.8395 private vs 0.8304 non-private, change -0.90 points $[-1.82, +0.02]$. No measurable cost. (Not read as "privacy helps": the interval covers zero.)
- **eICU:** $0.8245 \to 0.7936$, a cost of 3.09 points $[1.37, 4.81]$. Objective not met. AUPRC cost 1.72, balanced-accuracy cost 0.50 (met). End to end (pool-fitted constants): 3.91 $[2.40, 5.42]$; all three measures miss.
- **WEAR** (record-level): accuracy loss 1.97 pp $[1.76, 2.17]$; the 2-point threshold lies inside the interval, so it is inconclusive. Seeds were fixed in advance at 15 and not extended.

What it means. On eICU the cost comes from the smallest hospitals: with a dozen patients they must sample every patient at every step and carry the largest noise multipliers (up to 29.3).

5.5 Cost of the privacy unit: patient vs. record

Question. At the same budget, what does it cost to protect patients instead of records?
What we did. VitalDB, $\varepsilon = 1$, 5 paired seeds, record-level DP vs patient-level DP.
Result. AUROC $0.8520 \to 0.8373$, a cost of 0.0147 $[0.0042, 0.0253]$; AUPRC cost 0.0783 $[0.0532, 0.1035]$. For comparison, converting a record-level guarantee to patients by group privacy would turn $\varepsilon = 1$ into $\varepsilon \approx 8$; direct patient-level calibration instead needs only $1.77\times$ more noise.
What it means. Protecting the right unit has a real but moderate cost. Two causes: fewer units per step (4,372 patients vs 34,044 records), and clipping hurts patients with many records most. The k-law experiments isolate the second.

5.6 What makes strict privacy survivable: initialization

Question. Our recipe privatizes only the head over pretrained frozen encoders. Is the benefit from "head-only" DP or from pretraining?
What we did. Three arms, so the two effects can be separated, at six budgets from $\varepsilon = \infty$ to 0.5: **B4** full model, random initialization (the usual prior art); **B7** full model, pretrained initialization; **B6** head only, pretrained (ours).
Result. (VitalDB, patient-level, 3 seeds, after the fix.)

- B4 $\to$ B7 (pretraining): $+0.013$ at $\varepsilon = \infty$ rising to $+0.199$ at $\varepsilon = 0.5$; it grows at every one of the 5 steps.
- B7 $\to$ B6 (head-only): -0.003 to -0.006, every interval covers zero.

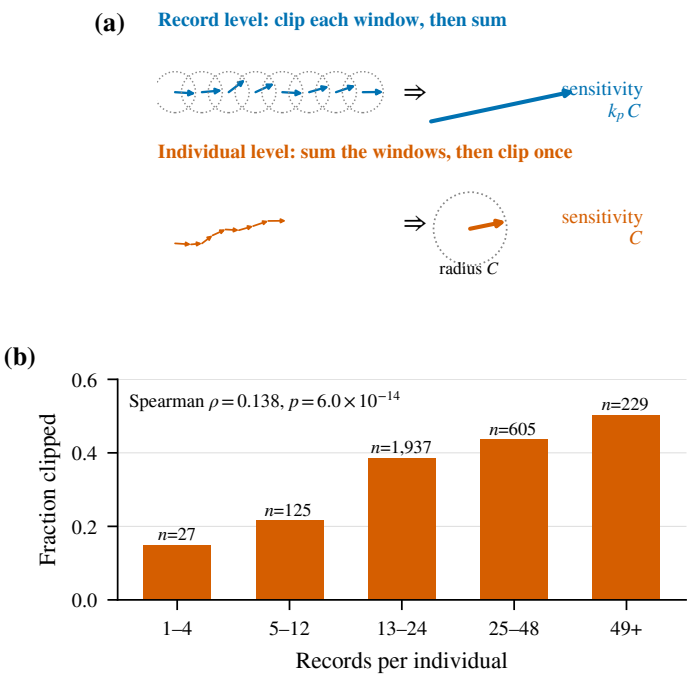


Figure 3: Record-level vs patient-level privacy unit.

- From $\varepsilon = \infty$ to 0.5 the random model loses 0.19 AUROC, the pretrained one 0.004.

What it means. Pretraining does the work; head-only DP adds nothing to accuracy. The effect grows exactly as privacy tightens and nearly vanishes without privacy, a built-in dose–response. The value of head-only training is *communication*: only the head is transmitted, which is where the 90 % uplink saving comes from. (Note: freezing does not reduce the noise; noise depends on budget, steps and sampling, never on parameter count.)

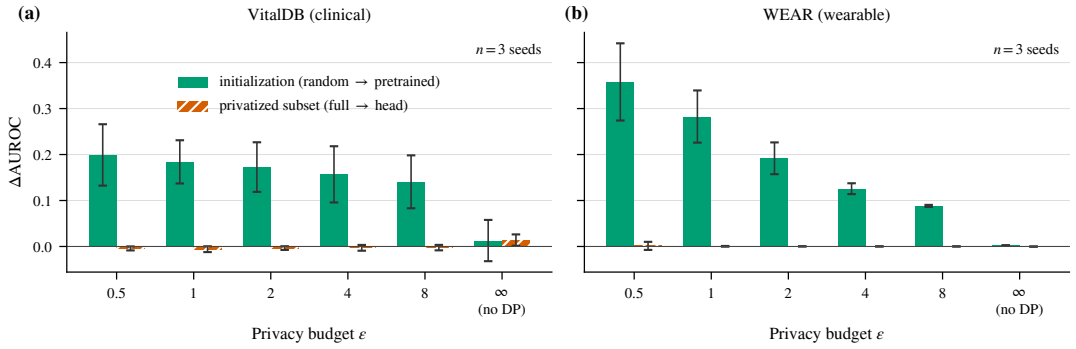


Figure 4: Initialization (B4→B7) vs privatized subset (B7→B6) across budgets.

5.7 Does the hierarchy help? (component ablation)

Question. What does each component (hierarchy, DP, sparsification, quantization) contribute?
What we did. Removed one component at a time; the hierarchy test was pre-registered and run at 6 seeds.
Result. Removing the hierarchy (one flat tier) changes AUROC by -0.0047 $[-0.0088, -0.0006]$, $p = 0.032$ on VitalDB, and -0.0052 $[-0.0074, -0.0030]$, $p = 0.0018$ on WEAR, at matched ε. Sparsification, quantization and loss-aware weighting are well-powered nulls on both corpora.
What it means. The hierarchy does not cost accuracy; removing it costs a small but resolved 0.005. Not claimed as "the hierarchy improves accuracy", because the flat arm also changes the number of local rounds. Compression is free.

5.8 Does multimodal fusion help?

Question. Does combining the two modalities beat the best single modality?
What we did. Fused model vs each single modality, 5 seeds; on eICU also a pre-registered "balance" test that degrades modality B with noise until the two modalities are equally strong.
Result.

- **VitalDB:** fused 0.8653 vs 0.8283, a gain of 0.0370 AUROC (5.9 seed SD) and 0.0671 AUPRC. Resolved.
- **WEAR:** one modality already reaches 0.9976, so there is no headroom to test.
- **eICU, natural:** no gain (+0.0037, within noise).
- **eICU, balanced** (amendment F-A1): fused 0.7282 vs 0.6992, gain +0.0291, positive on 5/5 seeds, paired $p = 0.019$ (control +0.0020).

What it means. Fusion pays when the second modality is competitive; when one modality dominates (eICU's APACHE score), it adds nothing. An early 2-seed figure (+0.0411) shrank at 5 seeds; only the 5-seed figure is quoted.

5.9 The k-law on VitalDB (records per patient)

Question. Does the cost of patient-level privacy grow with records per patient, k?
What we did. Built training sets in which every patient has exactly $k \in \{1, 2, 4, 8\}$ records, and trained a record-level arm and a patient-level arm with the same units per step. 3 seeds, AUPRC primary.
Result.

k	Gap (record − patient)	95 % half-width	p
1	−0.0071	0.0599	0.661
2	+0.0486	0.0129	0.004
4	+0.0796	0.0163	0.002
8	+0.0872	0.0248	0.004

The curve $A(1 - 1/k)$ with one parameter fits with $A = 0.1015$ (RMSE 0.0119); a two-parameter curve does no better (0.0117). Fitted on $k \leq 4$ only, it predicts 0.0904 at $k = 8$; observed 0.0872. The patient arm is flat (0.508–0.513); the record arm rises (0.502 → 0.596).
What it means. At $k = 1$ a patient is a record, so the two arms must agree; they do. That is a built-in positive control. The growing gap is the record arm exploiting extra records that the patient arm cannot, not the patient arm getting worse.

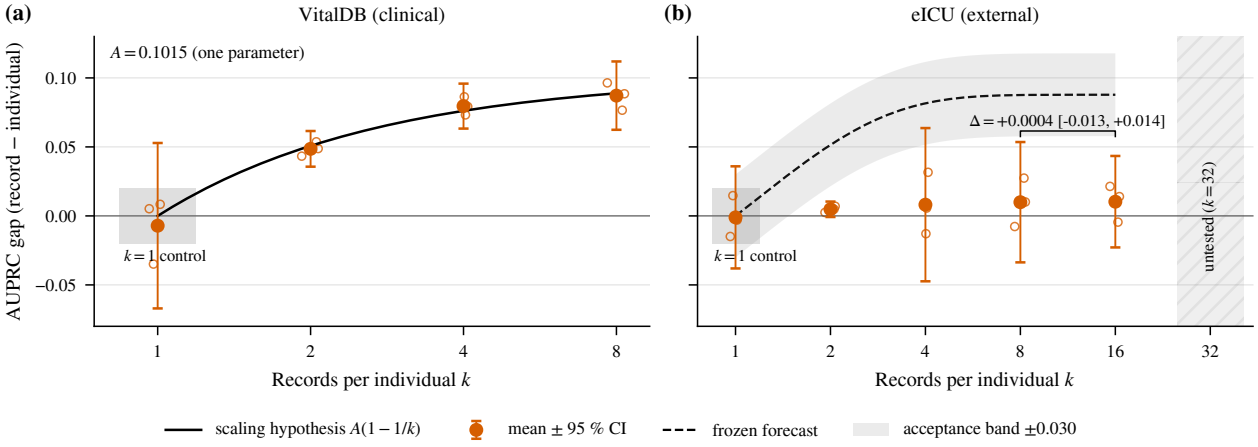


Figure 5: The k-law: (a) VitalDB with the fitted curve; (b) eICU against the frozen forecast.

5.10 External test of the k-law on eICU (pre-registered)

Question. Does the law transfer to a different hospital corpus?

What we did. Before the eICU corpus was built, we froze a forecast from VitalDB, $\text{gap}(k) = 0.0878\,(1-e^{-0.883(k-1)})$, with an acceptance band of ± 0.030, and extended k to 16 to test saturation. ($k = 32$ was dropped by a dated amendment before any result past $k = 8$: only 20 % of eICU patients reach 32 records.)
Result. (after the fix, 3 seeds)

k	Gap	Forecast	p
1	-0.0011 ± 0.0369	0.0000	0.909
2	$+0.0048 \pm 0.0057$	0.0515	0.068
4	$+0.0081 \pm 0.0554$	0.0816	0.593
8	$+0.0099 \pm 0.0437$	0.0876	0.431
16	$+0.0103 \pm 0.0332$	0.0878	0.314

From $k = 8$ to 16 the gap changes by $+0.0004$ $[-0.0134, +0.0141]$.
What it means. Shape transferred, size did not. The control holds, the gap rises and then flattens, as forecast. But at $k = 8$ it is about one ninth of the forecast, and no gap is significant at three seeds (about 25 seeds would be needed). One hypothesis: eICU's attainable precision is about half of VitalDB's (AUPRC 0.22 vs 0.50 at 4.2 % prevalence). Distinguishing that from a corpus-specific ceiling needs a third corpus. Seeds are *not* being added to chase significance.

5.11 Is the eICU shortfall just the budget? (matched grid)

Question. The forecast was fitted at $\varepsilon = 1$, batch 32; eICU ran at 0.8, batch 64. Is that why eICU is smaller?
What we did. Re-ran the VitalDB k-grid at the eICU operating point.
Result. The VitalDB gaps do not shrink ($+0.0914$ vs $+0.0872$ at $k = 8$). But this grid fails its own $k = 1$ control ($+0.0398$, all seeds the same sign), probably from the smallest pool combined with the largest batch. Correcting every grid for its own $k = 1$ offset attributes about half (51 %) of the $k = 8$ difference to the operating point.
What it means. Under either reading the budget does not explain the factor of nine, and the corpus effect never changes sign.

5.12 Is the cost an artefact of the clipping rule?

Question. Clipping a patient's *summed* gradient hurts patients with many records. Would normalizing by record count before clipping recover the cost? (It changes nothing about the privacy guarantee.)
What we did. Three rules: fixed bound, divide by k, divide by $\sqrt{k}$. Pre-registered criterion: count normalization must recover at least $+0.010$ AUPRC.
Result.

- **eICU, before the fix:** change -0.0001; the one-sided upper bound $+0.0058$ excludes $+0.010$. Fails.
- **VitalDB at $k = 8$, after the fix (D1):** clipped fraction falls from 0.50 to 0.34, AUPRC changes by $+0.0038$ $[-0.0072, +0.0148]$; the upper bound $+0.0113$ does *not* exclude $+0.010$: **inconclusive** by its own rule, though even the bound is one eighth of the 0.0914 cost.
- **eICU, after the fix (D2):** final-round clipped fraction falls from 0.14 to 0.08, AUPRC changes by -0.0012 $[-0.0093, +0.0069]$; the upper bound $+0.0043$ excludes $+0.010$. **Fails** (as before the fix). The square-root rule changes AUPRC by -0.0031.

What it means. Normalization changes how often clipping binds but not accuracy. On eICU the clipping rule does not explain the cost; on VitalDB it could explain at most one eighth of it. The cost belongs mainly to the patient unit itself.

5.13 Privacy audit: membership inference (LiRA)

Question. Can an attacker tell whether a *patient* was in training, and does patient-level DP leak less than record-level DP?
What we did. LiRA attack with 16 shadow models per arm, 5 target seeds, 4 attacks. A positive control (a no-DP model known to memorize) had to pass first.
Result. (after the fix, stage E2) Positive control AUC 0.623, 0.060 above its difficulty floor: the attack works. Record-level DP 0.548, patient-level DP 0.524 (only 0.008 above its floor). Record minus patient, paired over 5 seeds: loss $+0.012$ $[-0.006, +0.030]$, offline LiRA $+0.029$ $[-0.038, +0.095]$, online LiRA -0.003 $[-0.015, +0.009]$; online LiRA with global variance -0.029 $[-0.048, -0.010]$ ($p = 0.014$), i.e. the record arm leaks *less*, which does not survive correction for four attacks.
What it means. The large predicted difference (0.7 vs 0.5) is refuted. A small difference is not excluded at this sample size. The attack works (control passes), so the null is meaningful.

5.14 Private conformal prediction (negative result)

Question. Can each client release a private calibration threshold so that the system outputs prediction sets with 90 % coverage?
What we did. Exponential-mechanism quantile per client ($\varepsilon_{\text{conf}} = 0.1$), aggregated by edges and cloud; coverage measured on the test split.
Result. Thresholds land mid-range (0.42–0.61). The private model's probabilities sit near zero, so every set is just the majority class: coverage equals the base rate (80.1 % on VitalDB, fails; 95.7 % on eICU), and *no positive patient is covered*.
What it means. With about 100 calibration patients per client at $\varepsilon = 0.1$, the mechanism cannot separate thresholds. Conformal prediction was therefore removed from the system and the contributions, and the full budget $\varepsilon = 0.9$ goes to training. It stays in the paper as an honest negative result.

5.15 Standardization constants: a hidden release

Question. Input normalization means and SDs are fitted on training patients. Do they leak?
What we did. Measured, exactly, how much one patient can move them (leave-one-out over 92,909 eICU individuals), and compared them with constants fitted on the public pool.
Result. Removing one patient moves any mean by at most 0.0017 SD; for 99 % of patients any SD by at most 0.02 %; but one outlier moves one channel's SD by 5.2 %. Pool-fitted constants differ by at most 0.026 SD and 4.4 %.
What it means. Small, but it is a release outside the guarantee. Fix: re-standardize eICU with pool-fitted constants (verified exact: the stored data reproduce the original constants to 4×10^{-8}), re-pretrain the encoders and re-run. That run is the only configuration that is private end to end.
Result. AUROC 0.8209 $\rightarrow$ 0.7819, a cost of 3.91 points $[2.40, 5.42]$ ($p = 0.002$, 5 seeds); AUPRC 2.66 $[1.82, 3.49]$; balanced accuracy 2.69 $[1.57, 3.82]$.
What it means. Public constants make the guarantee end to end and cost about 0.8 AUROC points more than training-fitted ones (3.09); now all three measures miss the 2-point target.
Status: done 29 Oct.

5.16 Cost of disjoint encoders on VitalDB (F2)

Question. How much does it cost to pretrain VitalDB's encoders on a disjoint pool instead of on client data?
What we did. Stage F2: the headline patient-level run on VitalDB repeated with encoders pretrained only on the disjoint pool, five seeds, private and non-private arms.
Result. Against the oracle encoders the private model loses 0.0157 ± 0.0046 AUROC (0.8395 $\rightarrow$ 0.8237, $p < 0.001$) and 0.0122 ± 0.0262 AUPRC (not resolved). With disjoint encoders the cost of privacy itself is $+0.0014$ $[-0.0033, +0.0062]$ AUROC and $+0.0221$ $[-0.0003, +0.0446]$ AUPRC.

(Before the fix, record-level: 0.0322 AUROC, 0.0763 AUPRC.)
What it means. Honest encoders cost about 1.6 AUROC points on VitalDB, half the earlier estimate. Standardization constants are still fitted on client data here, so this run is not end to end; only the eICU re-run with public constants is.

6 Mistakes found and fixed

A large part of this project was finding errors that produced plausible-but-wrong numbers without any crash. Each one was fixed, tested, and the affected results re-run. The important ones:

Error	Effect and fix
Noise calibrated from one client	Every DP arm exceeded its budget. Now calibrated against all clients.
Conformal calibration fired 5×	Total ε was 3.49, not 0.96. Fixed to fire once; later removed entirely.
NS-3 had never run	Every early "network result" was an analytical fallback. NS-3 built; packet-level every round.
Stale partition metadata	Noise calibrated on wrong counts. Rebuilt.
eICU training collapse	Model decayed to constant output in long epochs, which would have poisoned every eICU k-number. Fixed validation cadence.
Divided by the realized batch $\lvert S\rvert$ (22 Oct)	Not the analysed mechanism; at small hospitals it reveals the patient count. Now divides by the expected batch. All patient-level results re-run in stages A–G.
Headline runs were record-level	The system runs used record-level DP while claiming patient-level. Re-run at patient level.
Aggregation weights leaked sizes	Now static declared sizes.

Why this matters. Every one of these would have survived review unnoticed and would have broken the central claim. The paper's numbers are all from the corrected code; the old results are archived, not overwritten.

7 Honest scorecard: what is and is not claimed

Claim	Status
Patient-level guarantee for hierarchical multimodal FL, all tiers accounted	proven (Theorem 1)
Communication, latency, energy objectives	met on all three corpora
No measurable utility cost on VitalDB at $\varepsilon = 0.9$	supported
Utility cost ≤ 2 points on eICU	**not met** (3.09; 3.91 end to end)
Pretrained initialization makes strict privacy survivable	strongly supported
Head-only DP improves accuracy	**refuted** (it helps communication)
k-law shape on VitalDB	supported, with positive control
k-law transfers to eICU	shape yes, magnitude no, unresolved
Cost is not a clipping artefact	eICU yes; VitalDB inconclusive ($\leq 1/8$ of cost)
Patient DP leaks much less than record DP (MIA)	large effect refuted
Private conformal prediction	**failed**, removed
Fusion helps	VitalDB yes; eICU only when balanced
Hierarchy costs accuracy	no (removing it costs 0.005)

None of the building blocks is new (user-level DP, private fine-tuning, hierarchical FL). The contribution is their composition inside a hierarchy with a proof, and careful *measurement* of what it costs.

8 Still to do

Item	What it gives	Expected
HED-FL, FL2DP baselines	empirical comparison	needs the two PDFs
Final consistency pass	paper ready to submit	after numbers freeze

Your decisions: the journal (IEEE IoT Journal or TII), authors, bios, acknowledgment, and a final reference check.

9 Glossary

Federated learning (FL) Training where data stays at each site and only model updates are shared.

Hierarchical FL FL with an intermediate edge tier that aggregates nearby clients before the cloud.

Differential privacy (ε, δ) A guarantee that the output barely changes whether or not one unit's data is included. Smaller ε is stronger; $\varepsilon \leq 1$ is considered strong.

Privacy unit What the guarantee protects: one record, or one patient.

DP-SGD Training with per-unit gradient clipping and Gaussian noise.

Clipping norm C The maximum size of one unit's contribution.

Noise multiplier σ Noise scale relative to C.

Accountant (PRV, RDP) Computes the total ε over all steps.

AUROC / AUPRC Ranking quality of the model; AUPRC is more sensitive when positives are rare (eICU: 4.2 %).

Seed One independent repetition of a training run; paired seeds compare two arms on identical splits.

Pre-registration Writing down the prediction and the pass/fail rule before running the experiment, so the result cannot be reinterpreted later.

LiRA Likelihood-ratio membership-inference attack using shadow models.

Oracle vs disjoint encoders Pretrained on client data (upper bound, not private) vs on a separate public pool (private end to end).